

Digital Twin for Urban Mobility Systems

Preliminary Research Documentation

Project Description: To develop an AI-driven Digital Twin for urban mobility that can simulate, predict, and optimize transportation systems by integrating traffic, public transit, shared mobility, EV infrastructure, and connected/autonomous vehicles to improve efficiency, safety, resilience, and sustainability.

1. Introduction

Urban transportation systems are becoming increasingly complex. A city can have roads, cars, buses, trains, taxis, ride-sharing vehicles, parking systems, electric vehicles and charging stations operating at the same time. These systems generate large amounts of data from GPS, traffic sensors, public transport, vehicles, parking systems and environmental information.

The main challenge is not only collecting this data but using it together to understand the current transportation situation, predict future conditions and support better transportation decisions.

A Digital Twin provides a virtual representation of a real transportation system. It can be used for understanding current conditions, prediction, scenario simulation and decision support.

2. What is a Digital Twin?

A Digital Twin is a digital representation of a real-world system that uses information from the real system to understand, simulate and analyze its behavior.

For urban mobility, a Digital Twin can represent:

- Roads
- Vehicles
- Traffic
- Public transport
- Parking
- Shared mobility
- Electric vehicles
- Charging stations
- Passenger demand

The basic process is:

Real transportation system → Data → Digital Twin → Prediction/Simulation → Decision/Optimization.

3. Project Motivation

The reviewed research shows that Digital Twins are already being used for different transportation problems, including parking and mobility forecasting, traffic congestion and traffic control, and autonomous transportation and predictive maintenance.

The mentor's project document proposes broader directions involving multimodal transportation, ETA prediction, ride-sharing, EV charging, vehicle relocation, fleet dispatch and DTUMOS scalability.

4. Research Paper 1

Title: [A digital twin framework for urban parking management and mobility forecasting](#)

Authors: Piccialli et al. | Journal: Nature Communications | Year: 2025 | Location: Caserta, Italy

4.1 Problem Statement

The paper discusses problems caused by increasing urbanization, including traffic congestion, inefficient transportation, pollution and increasing complexity of urban mobility. The authors explain that many existing Digital Twin systems focus on individual subsystems rather than understanding relationships between different components of urban mobility.

4.2 Objective

The Digital Twin has three main functions:

1. Descriptive analysis
2. Predictive forecasting
3. Generative or what-if simulation

4.3 Data Used

- Parking meter transactions
- Parking revenue
- Street occupancy
- Parking violations
- Parking sensor data
- Weather
- Air quality
- Points of Interest
- Events
- Agent shift schedules

4.4 Methods Used

- Spatial-Temporal Identity (STID) for prediction
- CVAE-WGAN for generative what-if simulation
- Linear Programming for automated agent shift scheduling
- Spearman correlation, Granger causality, Mann-Kendall test and ADF stationarity test

4.5 Main Contribution

The system combines data analysis, prediction and what-if scenario generation for parking and local mobility. It can investigate scenarios such as road closures, new parking facilities and rainy days.

4.6 Limitations

- Data quality issues, including limited granular environmental and user-behavior information.
- Sensor failures are partly handled through filtering and imputation, while more advanced anomaly detection remains an opportunity.
- Model results are not always easy for non-technical stakeholders to interpret.
- The system is mainly a decision-support system with human involvement rather than a completely autonomous cyber-physical control system.
- The prediction system is not continuously retrained, creating possible model-drift concerns.
- The main focus is parking and local mobility rather than a complete urban transportation system.
- Validation is based on Caserta data, so wider validation is needed for other cities and larger urban environments.

5. Research Paper 2

Title: [Green Transportation Planning for Smart Cities: Digital Twins and Real-Time Traffic Optimization in Urban Mobility Networks](#)

Authors: Marek Lis and Maksymilian Mądział | Journal: Applied Sciences | Year: 2026 | Location: Pobitno Roundabout, Rzeszów, Poland

5.1 Problem Statement

The paper focuses on traffic congestion at an urban roundabout. Congestion causes increased delay, stop-and-go traffic, reduced traffic efficiency and environmental impacts. The authors also discuss the difference between Digital Twins with active interaction/control and systems focused mainly on visualization or monitoring.

5.2 Objective

The objective is to combine Digital Twin technology, traffic microsimulation, real-time traffic information and traffic control to investigate dynamic traffic management.

5.3 Methodology

- PTV Vissim for traffic microsimulation
- GEH statistic for model calibration and validation
- Adaptive Inflow Metering (AIM) for traffic control
- Comparison of conventional roundabout, turbo-roundabout and flyover

5.4 Important Finding

The paper shows that changing physical infrastructure does not automatically solve congestion. The turbo-roundabout performed poorly under the tested peak traffic conditions, supporting the importance of dynamic traffic management in addition to infrastructure changes.

5.5 Limitations

- The AIM control system was tested through simulation/Software-in-the-Loop rather than deployment on physical traffic infrastructure.
- The study focuses on one roundabout in Rzeszów.
- Detailed turn-movement information was not available.
- Control thresholds were not extensively field-tuned.
- Real-world implementation would require reliable sensors, communication, data fusion and cybersecurity.
- The study does not establish performance across different cities and intersection types.
- The control strategy uses predefined rules rather than a learning-based control system.

6. Research Paper 3

Title: [Enhancing Smart Urban Mobility through Digital Twin-driven Autonomous Transportation and Predictive Maintenance](#)

Authors: Michael Oluwaseyi Aderibigbe et al. | Year: 2025

6.1 Problem Statement

The paper discusses transportation systems that are often reactive, fragmented and difficult to manage in real time. It focuses on connecting Digital Twin technology with autonomous transportation, traffic optimization and predictive maintenance.

6.2 Objective

The objective is to develop a Digital Twin-driven framework that supports real-time transportation decisions, traffic optimization, predictive maintenance and autonomous transportation management.

6.3 General Architecture

Physical transportation system → Sensors/Data → Digital Twin → AI/Optimization → Decision

6.4 AI/ML Components

- Machine Learning
- Deep Learning
- Reinforcement Learning
- Predictive maintenance
- Traffic optimization

6.5 Main Contributions

The paper combines Digital Twin, traffic management, autonomous transportation and predictive maintenance to move beyond simple monitoring toward prediction and decision-making.

6.6 Limitations and Challenges

- Different sensors and systems generate different types of data, creating data-integration challenges.
- Large transportation systems require significant computational resources.
- Different systems and Digital Twin platforms need interoperability.
- Vehicle and location data create privacy concerns.
- Real-world implementation requires suitable communication and computing infrastructure.
- Moving from framework/simulation toward large-scale real-world deployment remains challenging.

7. Comparison of the Three Papers

Area	Paper 1	Paper 2	Paper 3
Main focus	Parking	Traffic	Autonomous transport
Prediction	Yes	Yes	Yes
Simulation	Yes	Yes	Yes
What-if scenarios	Yes	Limited	Discussed
Traffic control	Limited	Yes	Yes
Predictive maintenance	No	No	Yes
Multimodal transportation	Limited	Limited	Broader framework
Data integration	Challenge	Challenge	Challenge
Scalability	Challenge	Challenge	Challenge
Real-world deployment	Limited	Not demonstrated	Challenge
Multiple-city validation	Limited	Limited	Limited

8. Common Research Gaps

Fragmented transportation systems: Existing work often concentrates on a specific area such as parking, traffic or maintenance instead of integrating several transportation components.

Data integration: Urban mobility uses GPS, traffic, public transport, vehicle, parking and other data sources with different formats, frequencies and quality.

Prediction versus decision-making: Prediction alone is not enough. A useful system should also support simulation, decisions and optimization.

Real-time operation: Moving from simulation or decision support toward continuously updated real-world operation remains challenging.

Scalability: More vehicles, sensors, data sources and simulations increase computational requirements.

Multimodal transportation: The reviewed papers do not provide a complete solution covering road traffic, public transit, shared mobility, EV infrastructure and connected/autonomous vehicles together.

9. Proposed Research Gap

Existing Digital Twin research has demonstrated applications in parking management, traffic optimization, mobility forecasting and predictive maintenance. However, these studies generally focus on specific transportation components or locations. There is a need to investigate how heterogeneous mobility data can be integrated within a Digital Twin to support prediction, multimodal transportation simulation and coordinated mobility decision-making.

10. Proposed Problem Statement

Urban transportation systems generate large amounts of data from vehicles, roads, public transport and shared mobility services. However, these data sources are often handled separately, making it difficult to accurately predict travel time and evaluate transportation decisions across different modes. Existing Digital Twin studies have demonstrated applications such as parking prediction, traffic optimization and autonomous transportation, but their scope is often focused on specific transportation components or individual locations. Therefore, this project aims to develop an AI-driven Digital Twin framework that integrates heterogeneous mobility data to predict travel time and simulate multimodal transportation scenarios, supporting improved routing and mobility decision-making.

11. Proposed Main Objective

To develop an AI-driven Digital Twin framework for urban mobility that integrates heterogeneous transportation data to predict travel time and simulate multimodal transportation scenarios for improved mobility decision-making.

12. Specific Objectives

4. To collect, clean and integrate relevant mobility data such as GPS trajectories, traffic information, road-network information and public-transit information.
5. To develop machine-learning/deep-learning models for predicting travel time.
6. To develop a routing approach that can consider multiple transportation modes.
7. To represent the transportation environment digitally and simulate different transportation scenarios.

8. To use prediction and simulation results to support better routing and transportation decisions.
9. To evaluate the proposed system using prediction accuracy, travel time, waiting time, route efficiency and computational performance.

13. Research Questions

10. How accurately can machine-learning models predict travel time using urban mobility data?
11. How can heterogeneous mobility data be integrated into an urban mobility Digital Twin?
12. How can ETA prediction support multimodal route planning?
13. How can a Digital Twin be used to simulate different urban transportation scenarios?
14. How does the proposed approach perform compared with conventional travel-time prediction and routing approaches?

14. Proposed Methodology

15. Data Collection — collect suitable GPS, traffic, road-network and public-transit data.
16. Data Preprocessing — clean data, handle missing values, process GPS and timestamps, and create useful features.
17. Exploratory Data Analysis — study trip distribution, travel time, distance, traffic patterns, time-of-day patterns and spatial patterns.
18. Baseline ETA Models — start with Linear Regression, Random Forest and XGBoost.
19. Deep Learning Models — investigate LSTM and GRU.
20. Advanced Models — depending on the data and DTUMOS requirements, investigate Transformer, GNN and spatio-temporal GNN approaches.
21. Digital Twin Integration — integrate the prediction model with DTUMOS or the selected Digital Twin environment.
22. Scenario Simulation — test normal traffic, increased traffic, road closure and transit-delay scenarios.
23. Optimization — depending on the final scope, investigate multimodal routing, ride-sharing, dynamic dispatch or EV fleet management.

15. Dataset

The mentor's document recommends GPS datasets for ETA modelling, including the Porto Taxi Dataset and Sungnam taxi GPS data. It also points to Uber DeepETA and Google DeepMind traffic prediction resources. The initial recommendation is to start with one dataset, preferably the Porto Taxi Dataset, and understand it completely before adding more datasets.

16. Model Development Strategy

Use a progressive approach:

- Level 1 — Baseline: Linear Regression, Random Forest, XGBoost.

- Level 2 — Deep Learning: LSTM, GRU.
- Level 3 — Advanced: Transformer, GNN, Spatio-temporal GNN.
- Study STGformer and TransETA as advanced/reference approaches because they are specifically mentioned in the mentor's document.

17. Evaluation Metrics

For ETA prediction:

- MAE — average prediction error.
- RMSE — gives more importance to larger errors.
- MAPE — error relative to actual travel time.

For routing:

- Total travel time
- Waiting time
- Route distance
- Detour
- Route efficiency

For the Digital Twin:

- Simulation accuracy
- Prediction accuracy
- Computational time
- Response time
- Scenario performance

18. Expected Contribution

- Integrate relevant mobility data.
- Predict travel time using AI.
- Support multimodal transportation.
- Simulate transportation scenarios using a Digital Twin.
- Provide information that can support mobility decisions.
- Evaluate the effect of different transportation conditions.

19. Possible Future Extensions

Ride-sharing: use ETA and routing information to match passengers travelling in similar directions.

EV Fleet Optimization: consider battery level, charging stations, passenger demand and vehicle location.

Reinforcement Learning: investigate dynamic vehicle dispatch, ride matching or traffic control.

20. Project Scope

Core scope: AI-based ETA prediction integrated with an urban mobility Digital Twin.

Supporting components: mobility data integration, GPS processing, multimodal routing, Digital Twin simulation and what-if scenarios.

Optional extensions: ride-sharing, EV charging, fleet dispatch, reinforcement learning and scalability improvements.

21. Immediate Action Plan

24. Study DTUMOS — understand its architecture, existing modules, input/output, simulation capabilities, existing ETA/routing functionality and limitations.
25. Create a literature review table containing problem, objective, dataset, methodology, results, limitations and research gap.
26. Understand the selected dataset, including columns, GPS format, timestamps, trip duration, distance and missing values.
27. Build a simple ETA baseline before trying advanced models.
28. Compare progressively from baseline ML to LSTM/GRU and then Transformer/GNN if appropriate.
29. Integrate the working model with DTUMOS.
30. Run what-if scenarios.
31. Add an optimization component only after prediction and simulation are working.

22. Final Working Project Definition

Working Title: AI-Based Multimodal ETA Prediction and Mobility Optimization using a Digital Twin for Urban Mobility Systems

Working Problem: Urban mobility data comes from multiple transportation sources and is often handled separately. This makes travel-time prediction and coordinated transportation planning difficult. Existing Digital Twin research has demonstrated solutions for individual areas such as parking, traffic control and predictive maintenance, but there is an opportunity to investigate a more integrated approach for multimodal mobility.

Working Objective: Develop an AI-driven Digital Twin framework that integrates mobility data, predicts travel time and supports multimodal transportation simulation and decision-making.

Core Technology: Machine Learning + ETA Prediction + Multimodal Routing + Digital Twin

Possible Advanced Technology: LSTM / Transformer / GNN / Spatio-temporal GNN

Platform: DTUMOS

Initial Dataset: Porto Taxi GPS dataset

Possible Extension: Ride-sharing or EV fleet optimization

23. Important Note

This document is a preliminary research plan. The final problem statement and research gap should be finalized only after studying DTUMOS itself. The mentor's document lists multiple possible directions and explicitly allows students to work on a part of the larger project.